

TECHNICAL REPORT 001

Kernellum Compiler and Physical-Evidence Pipeline

From quantized models to routed architecture search and board-ready validation

RESEARCH LANE

SYSTEM / COMPILER / PHYSICAL EVIDENCE

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System-level technical report

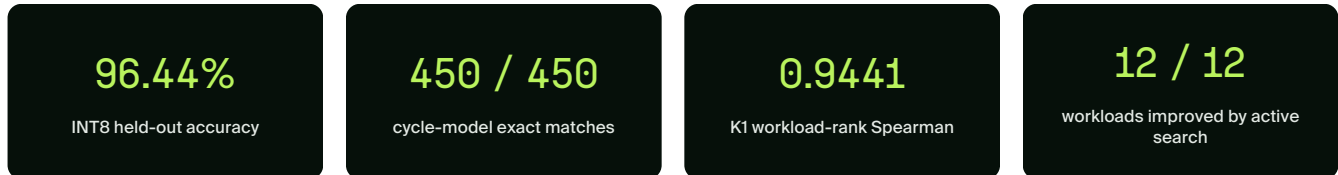
AUTHOR

Sushanth Dasari

EXECUTIVE SUMMARY

A research system that makes evidence strength explicit

Kernellum is an independent research program in workload-specific AI acceleration and physical-design-aware architecture search. The program couples model-to-RTL generation with synthesis, final-route measurement, bounded physical feedback, and a board-validation handoff so architecture decisions can be tested at progressively stronger evidence levels.



Evidence ladder

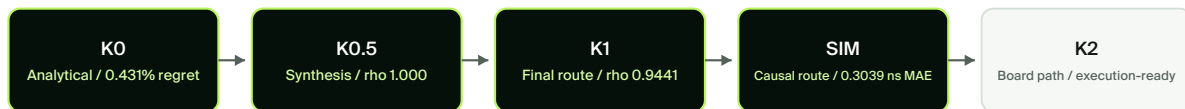


Figure 1. Unified evidence ladder from analytical search through final-route validation and the K2 board handoff.

01 / SYSTEM

Research architecture

The working principle is that analytical predictions remain useful only when implementation evidence is allowed to alter the next design decision. The resulting loop connects workload definition, architecture search, generated RTL, synthesis, final route, and physical-feedback selection.

Physical-evidence research loop

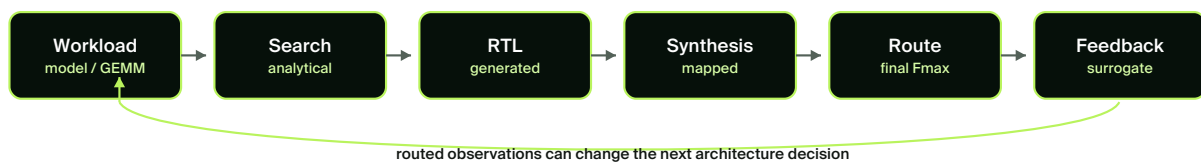


Figure 2. Kernellum research loop. Routed observations close the architecture-selection loop rather than acting as post-processing.

System architecture

The pipeline has three coordinated layers: functional correctness, implementation evidence, and decision logic. Each layer produces machine-readable artifacts that are carried forward into the next stage.

02 / MODEL TO RTL

Compiler evidence

The compiler demonstration uses a $64 \rightarrow 32 \rightarrow 16 \rightarrow 10$ multilayer perceptron on the scikit-learn handwritten-digits workload. Hidden layers use ReLU, the held-out set contains 450 samples, and the generated hardware path is verified independently from the floating-point model.

Metric	Observed
Floating-point held-out accuracy	96.22%
INT8 held-out accuracy	96.44%
Float / INT8 prediction agreement	99.78%
Independent cycle-model exact tensor matches	450 / 450
Generated RTL golden-vector simulations	32 / 32
Generic Yosys CHECK problems	0

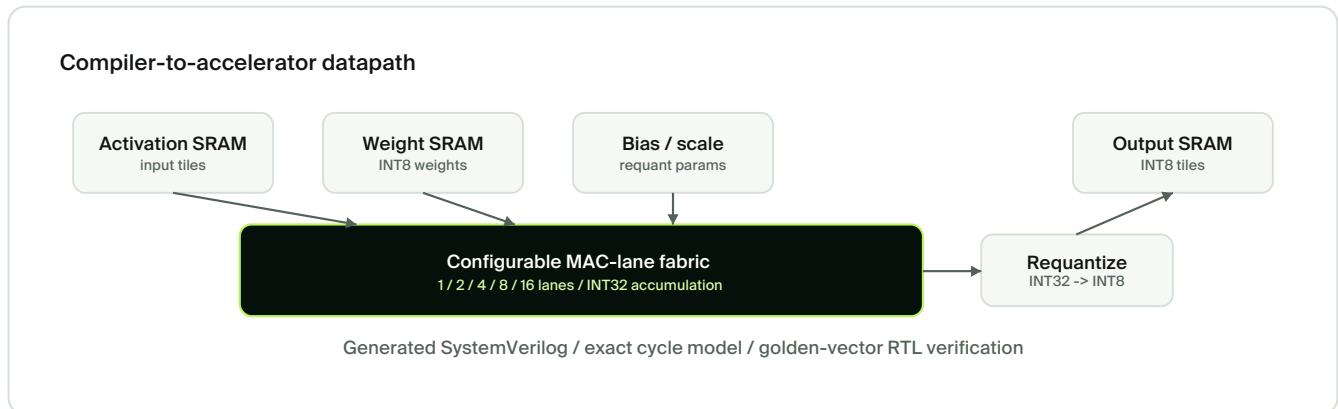


Figure 3. Generated accelerator datapath. Quantized memories, configurable MAC lanes, INT32 accumulation, and requantization form a deterministic model-to-RTL path.

03 / SEARCH

K0: analytical design-space search

K0 evaluates 12 dense GEMMs representative of a compact Transformer encoder at sequence lengths 64, 128, and 256. The frozen space contains 24,576 candidates per workload under fixed INT8 precision and hard DSP-equivalent and BRAM-equivalent limits. Exhaustive enumeration provides finite-space ground truth.

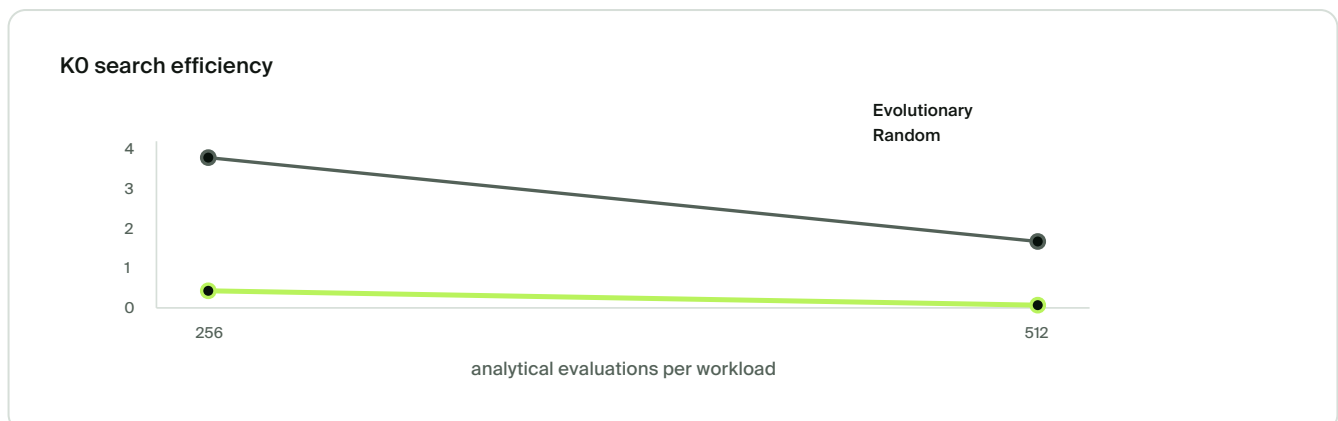


Figure 4. Analytical search efficiency. Evolutionary search reaches 0.431% mean regret at 256 evaluations and 0.069% at 512, substantially below the equal-budget random baseline.

04 / TECHNOLOGY MAPPING

K0.5: synthesis calibration

Nine stratified configurations are synthesized after functional simulation to test whether coarse architecture rankings survive elaboration and technology mapping.

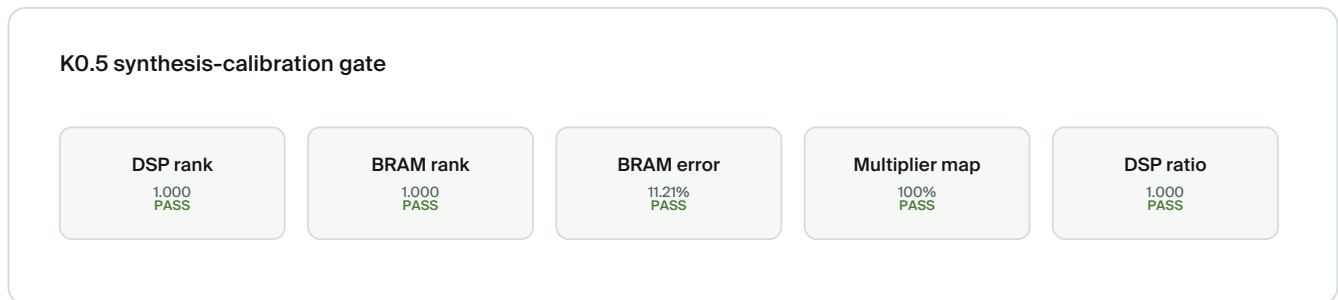


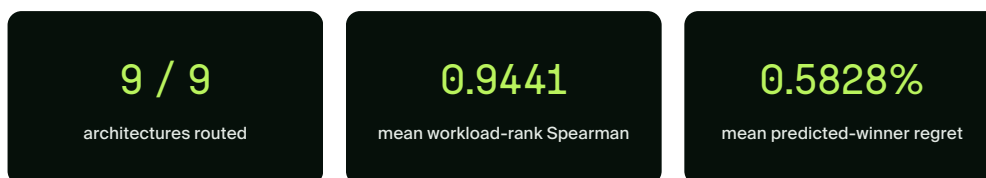
Figure 5. Preregistered K0.5 synthesis gate. All ranking, relative-error, multiplier-preservation, and DSP-mapping criteria passed.

Exact one-multiplier-per-PE preservation is retained across the sampled architecture family. DSP and BRAM ordering remain perfectly rank-correlated with the analytical model, while the observed BRAM granularity is captured explicitly for later stages.

05 / PHYSICAL IMPLEMENTATION

K1: final-route ranking

K1 replaces the fixed-frequency assumption with final-route ECP5 timing. Nine frozen architectures are synthesized and routed on the ECP5-85K target, with final timing sourced directly from nextpnr post-route reports.



Equal-PE orientation pairs and equal-geometry buffer-depth variants produce distinct final-route frequencies. Examples include 4x8 at 54.06 MHz versus 8x4 at 53.22 MHz, and 8x8 K_TILE 16 at 49.70 MHz versus K_TILE 64 at 45.83 MHz. These differences make physical layout decision-relevant.

06 / ACTIVE SEARCH

K1 closed-loop physical feedback

The closed-loop study starts from nine routed observations in a 172-architecture space. Four active acquisitions and a fixed-seed random arm receive equal route budgets. The full study physically attempts 17/172 designs, or 9.88% of the search pool.

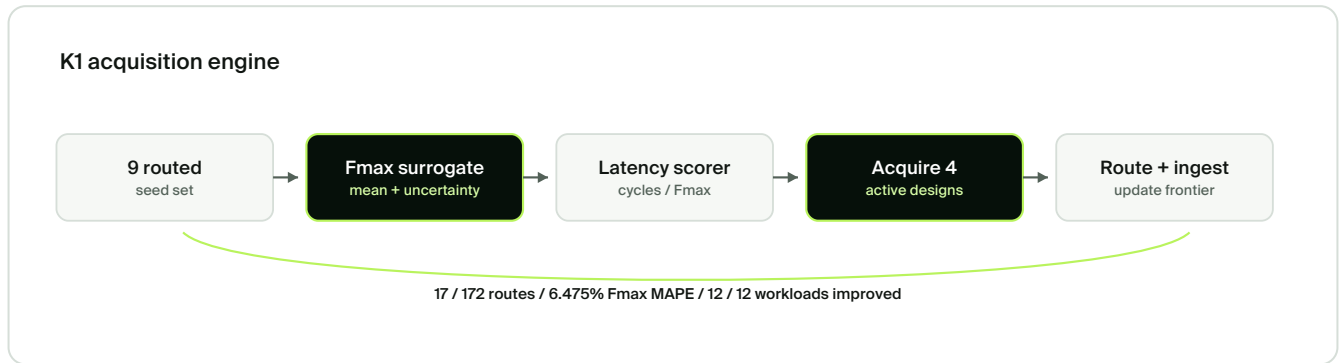


Figure 6. Closed-loop acquisition engine. A sparse routed seed set drives the next four implementation decisions through a surrogate-guided latency objective.

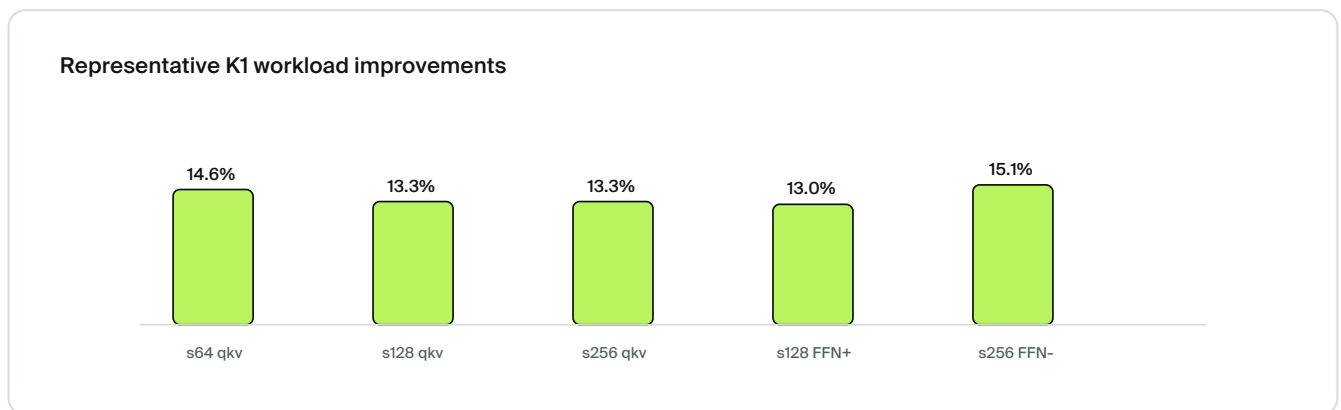


Figure 7. Representative workload improvements from the active routed search. The active arm improved all 12 Transformer GEMMs under the frozen budget.

Closed-loop result

All 8/8 new routes completed. New-candidate final-route-Fmax MAPE was 6.475%. Mean final-best latency reached 16.7398 ms for active search versus 19.4428 ms for the equal-budget random control, with 12/12 versus 0/12 workload improvements.

07 / CAUSAL ROUTING EVIDENCE

Project SIMILARITY: topology as a variable

Project SIMILARITY isolates operand-distribution topology under matched arithmetic fabrics. The corrected final-route experiment shows that direct broadcast and registered local transport scale differently with array size while preserving exact one-DSP-per-PE mapping.

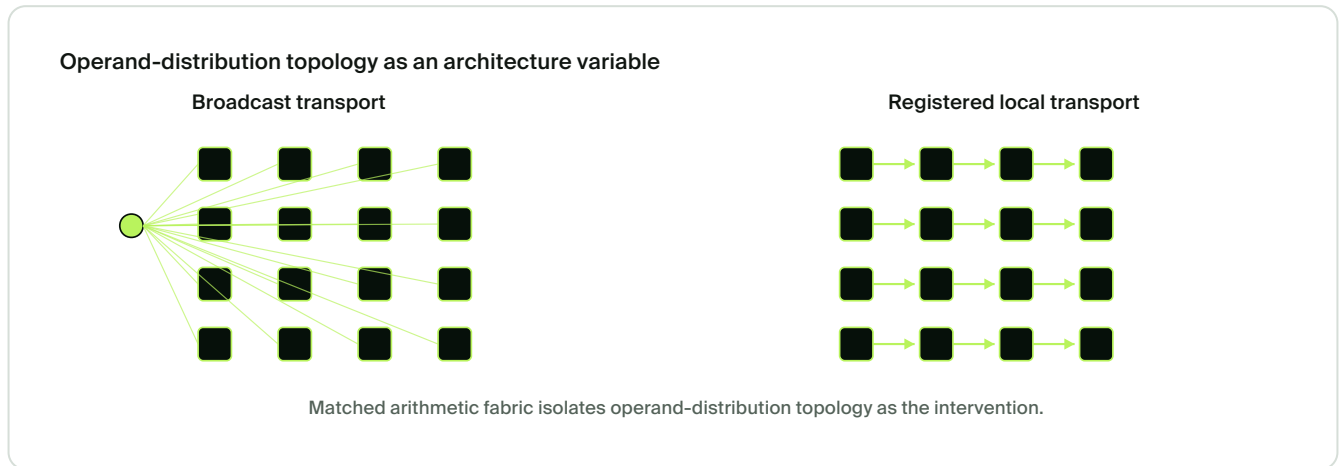


Figure 8. Causal topology intervention. Broadcast fanout is replaced by registered nearest-neighbor transport while arithmetic density is held fixed.

The corrected broadcast period slope is **0.4503 ns/ $\sqrt{\text{PE}}$** versus **0.1493 ns/ $\sqrt{\text{PE}}$** for local transport. A frozen differential law predicts unseen rectangular routing tax with **0.3039 ns** held-out MAE and **0.8921** correlation; the subsequent two-candidate portfolio beats the broadcast baseline in **36/36** unseen deployment cases.

O8 / MEASUREMENT STACK

K2 hardware-validation handoff

Three frozen ECP5 EVN variants have passed host tests, UART RTL simulation, synthesis, ECP5-85K place-and-route, the 12 MHz bring-up target, and bitstream generation.

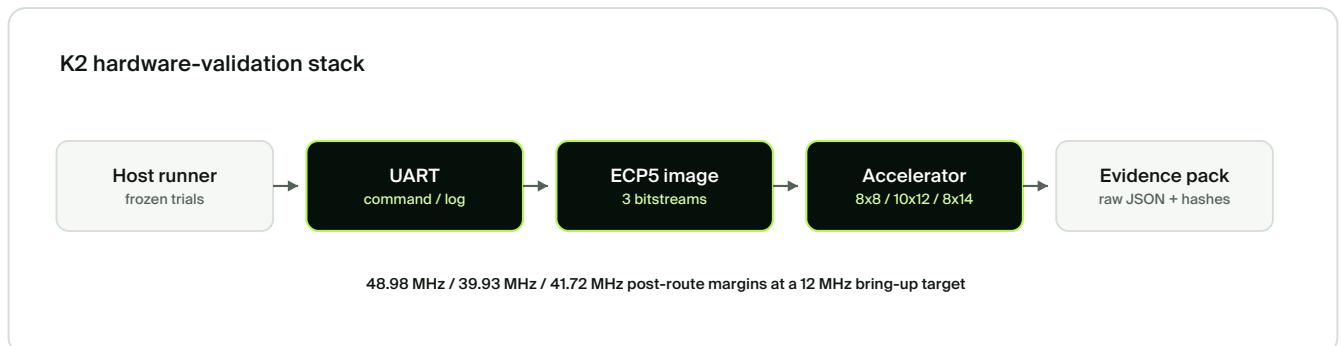


Figure 9. K2 board-validation stack. The host runner, UART protocol, frozen images, accelerator cores, and raw-log evidence pack define the next measurement release.

Variant	DSP	BRAM	LUT4	FF	Post-route Fmax
8x8 K32	64	4	8,928	2,551	48.98 MHz
10x12 K32	120	6	11,898	4,439	39.93 MHz
8x14 K64	112	6	15,055	4,188	41.72 MHz

Current evidence position

The compiler, synthesis, final-route ranking, closed-loop acquisition, and causal topology results are fully documented in the current public record. K2 packages the transition to physical board measurement as a separate evidence release.

Reproducibility record



Figure 10. Claim-to-artifact map. Each major public claim is tied to a named report layer and a machine-readable record family.

Stage	Primary public artifact
Compiler	artifacts/digits_int8/ / tests/test_pipeline.py
K0	docs/K0_REPORT.md / experiments/k0_transformer/run.py
K0.5	results/k05_synthesis.csv / results/k05_validation.json
K1	results/k1_routes.csv / results/k1_validation.json
K1 loop	results/k1_closed_loop_routes.csv / results/k1_closed_loop_validation.json
SIMILARITY	results/similarity_routed_law_summary.json / results/similarity_portfolio_summary.json
K2	results/k2_board_ready_manifest.json

Publication record

Kernellum Research Technical Report 001. Repository: github.com/sushxnthd/kernellum. Project site: sushxnthd.github.io/kernellum/.